

Positive completion of rational step data for two exceptional variables

Research proof: independently reviewed

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Status. The complete proof below has been independently reviewed by the root, size-bounds, and rational-design research agents.

Scope. This note constructs two finite equal-weight real random variables which, together with m independent continuous uniform variables, have all orders equally likely. It does not yet discretize the remaining m variables. In particular, it does not claim two polynomial-size exceptional dice in a completely finite family.

Theorem 1 (Positive completion). *Let $m \geq 1$, $0 < b_1 < \dots < b_K < 1$, and suppose the equal-weight rule at the b_q integrates every polynomial of degree at most $m + 1$ on $[0, 1]$. Write $b_0 = 0, b_{K+1} = 1$ and $\Delta = \min_{0 \leq q \leq K} (b_{q+1} - b_q)$. Suppose $0 = r_0 < r_1 < \dots < r_K < r_{K+1} = 1$ satisfy*

$$\frac{1}{K} \sum_{q=1}^K r_q b_q^j = \frac{1}{j+2}, \quad 0 \leq j \leq m. \quad (1)$$

There is an absolute $c > 0$ such that, if

$$\max_q |r_q - b_q| \leq c \frac{\Delta^{5/2}}{(m+1)^3}, \quad (2)$$

then a density f on $[0, 1]$, polynomial of degree at most m on each interval (b_q, b_{q+1}) , has the following properties:

1. $1/2 \leq f \leq 3/2$ almost everywhere;
2. $\int_0^{b_q} f = r_q$;
3. if A has density f , B is uniform on the K nodes b_q , and $U_1, \dots, U_m$ are independent uniform variables on $[0, 1]$, then all $(m+2)!$ strict orders of these independent variables have probability $1/(m+2)!$.

If all r_q are rational with a common denominator L , then, after multiplying L by an integer if necessary, A can instead be chosen uniform on L points in $\bigcup_q(b_q, b_{q+1})$, with the same conclusion. It is enough that every integer $L(r_{q+1} - r_q)$ be at least $C(m+1)^2$.

Define the tail step functions

$$S_j(a) = \frac{1}{K} \sum_{b_q > a} b_q^j, \quad 0 \leq j \leq m,$$

and the finite-dimensional space

$$V = \text{span}\{a^j : 0 \leq j \leq m\} + \text{span}\{a^r S_s(a) : r, s \geq 0, r + s \leq m\}.$$

Put $S = \text{span}\{1, S_0, \dots, S_m\} \subseteq V$. All norms and orthogonal projections below use Lebesgue measure on $[0, 1]$. Let P_0 be orthogonal projection onto the space of functions constant on each of the $K+1$ intervals (b_q, b_{q+1}) .

Lemma 1 (A stable quotient by the step functions). *For every $F \in V$,*

$$\text{dist}_{L^2}(F, S) \leq C(m+1)^2 \Delta^{-1} \|(I - P_0)F\|_2.$$

Moreover $V \cap \text{ran } P_0 = S$.

Proof. The jump of F at b_q is $-D(b_q)/K$ for a polynomial D of degree at most m . If $D(a) = \sum_j d_j a^j$, the function $G = \sum_j d_j S_j \in S$ has the same jumps. Consequently $F - G$ is a continuous piecewise polynomial on $[0, 1]$. Adding a constant to G , we may arrange $\int(F - G) = 0$. The ordinary Poincare inequality gives

$$\|F - G\|_2 \leq C\|F'\|_2.$$

On an interval I of length at least Δ , the polynomial inverse Markov inequality gives

$$\|F'\|_{L^2(I)} \leq C(m+1)^2 |I|^{-1} \|F - \text{mean}_I F\|_{L^2(I)}.$$

Summing its square over the cells proves the claimed bound. If $(I - P_0)F = 0$, the bound gives $F \in S$, proving the intersection claim. $\square$

Proof of Theorem 1: positive density. On (b_q, b_{q+1}) set

$$g = \frac{r_{q+1} - r_q}{b_{q+1} - b_q}.$$

Then $\int g = 1$, $\int_0^{b_q} g = r_q$, and

$$\|g - 1\|_\infty \leq \frac{2 \max_q |r_q - b_q|}{\Delta}. \quad (3)$$

The step equations imply

$$\int g S_j = \frac{1}{K} \sum_q b_q^j r_q = \frac{1}{j+2} = \frac{1}{K} \sum_q b_q^{j+1} = \int S_j,$$

so $g - 1$ is orthogonal to S . On $W = (I - P_0)V$ define

$$\Lambda((I - P_0)F) = \int (g - 1)F.$$

Lemma 1 makes this well defined and gives

$$\|\Lambda\| \leq C(m+1)^2 \Delta^{-1} \|g - 1\|_2.$$

The Riesz representative $h \in W$ therefore satisfies

$$\int h F = \int (g - 1)F \quad (F \in V), \quad \|h\|_2 \leq C(m+1)^2 \Delta^{-1} \|g - 1\|_2.$$

Here h has zero integral on each cell and is polynomial of degree at most m there. The polynomial L^2 - L^∞ estimate on each cell now yields

$$\|h\|_\infty \leq C(m+1) \Delta^{-1/2} \|h\|_2 \leq C(m+1)^3 \Delta^{-3/2} \|g - 1\|_2.$$

Set $f = g - h$. This retains every prescribed cell mass and satisfies $\int f F = \int F$ for all $F \in V$. Equations (2) and (3), with sufficiently small absolute c , give $1/2 \leq f \leq 3/2$.

For completeness, these tests imply the full order assertion. For $r + s \leq m$,

$$\mathbb{E}[A^r B^s \mathbf{1}_{A < B}] = \int f(a) a^r S_s(a) da = \frac{1}{K} \sum_q \frac{b_q^{r+s+1}}{r+1} = \frac{1}{(r+1)(r+s+2)}.$$

These are exactly the corresponding moments of two independent continuous uniforms. Ordinary moments of A and B through degree m are also correct. Subtracting the above triangle moments from their product moments gives the corresponding identities for $B < A$. For any prescribed order of $A, B, U_1, \dots, U_m$, conditioning on $A = a < B = b$ gives a polynomial proportional to $a^i (b - a)^j (1 - b)^k$, where $i + j + k = m$; the opposite triangle is identical with the labels exchanged. Thus every such probability equals its all-continuous uniform value $1/(m+2)!$. $\square$

Lemma 2 (Interior equal-weight quadrature). *Let a probability density w on $[0, 1]$ satisfy fixed bounds $0 < c_0 \leq w \leq C_0$. For every integer $N \geq C_{c_0, C_0}(m+1)^2$ there is a degree- m equal-weight quadrature for w whose nodes all lie in $(0, 1)$.*

Proof. Theorem 1.1 of Gilboa–Peled gives this bound with nodes in the closed interval for any density bounded above and below by fixed positive constants. We explain how to move the support strictly inside without changing the degree- m moments or these density bounds.

Take $J = [\tau, 1 - \tau]$, where $\tau = c(m+1)^{-2}$ and $c > 0$ is sufficiently small in terms of c_0, C_0 . Let $K_m^J(t, x)$ be the reproducing kernel for degree- m polynomials in $L^2(J)$. The usual Legendre kernel bound gives $|K_m^J(t, x)| \leq C(m+1)^2$ for $x, t \in J$. The same bound, with a changed absolute constant, holds for $x \in J, t \in [0, 1]$: the affine coordinate of t is at most $1 + C\tau$ in absolute value, and $|P_j(1+v)| \leq \exp(Cj\sqrt{v})$ for $0 \leq v \leq C\tau$. Define on J

$$k(x) = \int_{[0,1] \setminus J} w(t) K_m^J(t, x) dt.$$

Then $\|k\|_\infty \leq CC_0\tau(m+1)^2 \leq c_0/2$ and, for every polynomial P of degree at most m ,

$$\int_J P(x) k(x) dx = \int_{[0,1] \setminus J} P(t) w(t) dt.$$

The density $\tilde{w} = (w + k)\mathbf{1}_J$ has total mass one, agrees with w on these moments, and after affine rescaling of J has uniform positive lower and upper density bounds. Apply the closed interval theorem on J . Its nodes, including possible endpoints of J , belong to $(0, 1)$ as required. $\square$

Proof of Theorem 1: finite A. The mass of cell q is $\gamma_q = r_{q+1} - r_q = N_q/L$. Rescaling the conditional density f/γ_q from that cell to $[0, 1]$ gives density bounds $1/3$ and 3 , since f and its cell average both lie in $[1/2, 3/2]$. If $N_q \geq C(m+1)^2$, Lemma 2 gives a degree- m equal-weight quadrature with exactly N_q nodes strictly inside the cell. Give all $\sum_q N_q = L$ nodes mass $1/L$. This preserves the integral of every function which is polynomial of degree at most m in each cell, hence of every $F \in V$. The order argument above applies unchanged, and no A node coincides with a B node. $\square$

Corollary 1 (A polynomial-size conditional conclusion). *Suppose, as m varies, the step data in Theorem 1 can be chosen with K , Δ^{-1} , and a common denominator L all bounded by fixed powers of $m+1$, and with (2). Then two finite equal-weight independent variables with polynomially many atoms*

each can coexist with m independent continuous uniforms in a completely order-fair family.

Proof. Every cell mass is at least $\Delta/2$. Multiplying L by an integer at most $1 + C(m+1)^2/(L\Delta)$ makes all cell counts large enough. The resulting total count remains polynomial, and the theorem applies. $\square$

Lemma 3 (An alternative with prescribed truncated moments). *In the hypotheses of Theorem 1, put $d_q = r_q - b_q$, including $d_0 = d_{K+1} = 0$. For any integer $D \geq m$, if*

$$\max_q |d_q| \leq c\Delta/(D+1)^2,$$

there is a density f between $1/2$ and $3/2$, polynomial of degree at most D on each cell, such that

$$\int_0^{b_q} a^s f(a) da = \frac{b_q^{s+1}}{s+1} + d_q b_q^s, \quad 0 \leq s \leq D.$$

It yields the same hybrid order-fair family as in the theorem and may be discretized cellwise in the same way.

Proof. For $I_q = [b_q, b_{q+1}]$ let $K_D^{I_q}$ be its degree- D Lebesgue reproducing kernel. On that cell set

$$f(a) = 1 + d_{q+1} K_D^{I_q}(b_{q+1}, a) - d_q K_D^{I_q}(b_q, a).$$

The kernel is bounded in absolute value by $(D+1)^2/|I_q|$ on the closed cell. This proves positivity and, after telescoping the cell moment corrections, the asserted truncated moments. In particular the cell masses are the prescribed rational differences of ranks. The full moments of A are uniform because $d_{K+1} = 0$. For $r+s \leq m$,

$$\mathbb{E}[A^r B^s \mathbf{1}_{A < B}] = \frac{1}{K} \sum_q \left(\frac{b_q^{r+s+1}}{r+1} + d_q b_q^{r+s} \right) = \frac{1}{(r+1)(r+s+2)},$$

since (1) makes $K^{-1} \sum_q d_q b_q^j = 0$ for $j \leq m$. The triangle argument already proved finishes the claim. $\square$

Input and remaining gap. The companion construction by the rational-design research agent produces the step data with polynomial separation and denominator, using a small Legendre perturbation followed by a quantitatively conditioned implicit-function argument. This note isolates the positive completion so that its conditioning can be independently checked. An additional argument is required to replace the m continuous uniforms by finite dice while retaining the two small atom counts. The one-exceptional construction cannot simply be invoked: two distinguished cut points permit surviving two-bump interaction terms. That additional argument is now supplied by the independently reviewed `../size_bounds/two_polynomial_exceptional_dice.tex` and its clustered lifting companion. It uses rational box smoothing and preserves all mixed boundary terms explicitly; the scope of the present theorem itself remains the stated hybrid model. Lemma 3 is a later elementary strengthening, independently checkable from the reproducing-kernel identity; its special truncated moment formula is not needed for the quotient argument above.

References

- [1] S. Gilboa and R. Peled, *Chebyshev-type quadratures for doubling weights*, Constructive Approximation 45 (2017), 193–216, Theorem 1.1. arXiv:1507.01505.